Algoritmul QR cu deplasare

Alg QR cu deplasare este un alg eficient de determinare a valorilor proprii în cazul matricilor tridiagonale simetrice și în cazul matricilor Hessenberg

Ex:

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

$\begin{bmatrix} a_{n-1} & b_n \\ b_n & a_n \end{bmatrix}$ și calc val proprii

$$\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \Rightarrow \begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = 0 \Rightarrow \begin{cases} \lambda_1 = 2 \\ \lambda_2 = 4 \end{cases}$$

Selectăm cea mai apropiată val proprie de a_n // a_3

$$\begin{bmatrix} a_1^{(1)} & b_2^{(1)} & 0 \\ b_2^{(1)} & a_2^{(1)} & b_3^{(1)} \\ 0 & b_3^{(1)} & a_3^{(1)} \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

Alegem $\tau_1 = 2$ deoarece sunt la fel de depărtate de $a_n = a_3 = 3$

$$B = A - \tau_1 I_3 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Aducem B la forma sup Δ cu rotări Givens

$$G_1 = \begin{bmatrix} c & s & 0 \\ -s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad G_1 B = \begin{bmatrix} 0 & & \end{bmatrix}$$

$$-s + c = 0 \Rightarrow c = s = \frac{1}{\sqrt{2}}$$

$$G_1 B = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} \sqrt{2} & \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 1 \end{bmatrix}$$

$$G_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & s \\ 0 & -s & c \end{bmatrix}$$

$$G_2 \cdot G_1 B = \begin{pmatrix} x & x & x \\ x & x & x \\ x & 0 & x \end{pmatrix}$$

$$c = 0 \Rightarrow s = 1$$

$$G_2 \cdot G_1 B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{2} & \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 1 \end{bmatrix} =$$

$$= \begin{bmatrix} \sqrt{2} & \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & 1 & 1 \\ 0 & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} = R_1$$

$$B = G_1^T G_2^T R_1$$

$$A_1 = R_1 Q_1 = \begin{bmatrix} \sqrt{2} & \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & 1 & 1 \\ 0 & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \end{bmatrix}$$

$$G_1^T \Theta_2^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} =$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 1 & -\frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & 0 \end{bmatrix} \qquad \lambda_3^{(1)} \simeq 0$$

$$\det(A - \lambda I_3) = 0 \Rightarrow \begin{vmatrix} 2-\lambda & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 1-\lambda & -\frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda)\left[(-\lambda)(1-\lambda) - \frac{1}{2}\right] - \frac{1}{\sqrt{2}}\left[-\frac{1}{\sqrt{2}}\lambda\right] =$$

$$= (2-\lambda)\left(\lambda^2 - \lambda - \frac{1}{2}\right) + \frac{1}{2}\lambda = 0$$

Cel mai apropiat λ este $\frac{1}{2} - \frac{1}{2}\sqrt{3}$

$$\nu_2 = \frac{1}{2} - \frac{1}{2}\sqrt{3}$$

$$B = A_1 - T_2 \, i_3 = \begin{bmatrix} 2 - \frac{1}{2} + \frac{1}{2}\sqrt{3} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 1 - \frac{1}{2} + \frac{1}{2}\sqrt{3} & -\frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & -\frac{1}{2} - \frac{1}{2}\sqrt{3} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{2} + \frac{\sqrt{3}}{2} & \frac{1}{\sqrt{2}} & 0 \\ \boxed{\frac{1}{\sqrt{2}}} & \frac{1}{2} + \frac{\sqrt{3}}{2} & -\frac{1}{\sqrt{2}} \\ 0 & \boxed{-\frac{1}{\sqrt{2}}} & -\frac{1}{2} - \frac{\sqrt{3}}{2} \end{bmatrix}$$

$\downarrow$ G_1 $\qquad$ $\rightarrow G_2$ la $G_1 B$

$G_2 \cdot G_1 \cdot B = R_2 \Rightarrow B = G_1^T G_2^T R_2$

$A_2 = R_2 (G_1^T G_2^T) = R_2 Q_2 =$

$$= \begin{bmatrix} 2,672 & 0,375 & 0 \\ 0,375 & 1,473 & 0,03 \\ 0 & \underset{\approx 0}{\boxed{0,03}} & -0,047 \end{bmatrix} \Rightarrow \lambda_3^{(2)} \simeq -0,047$$

λ_1, λ_2 se det direct

Precizări

* Scopul alg este de a aplica transformări asupra matricii A a.î. b_2 sau $b_n \to 0$, situație în care putem aproxima λ ca fiind a_1, respectiv a_n.
* Este la alegerea noastră să lucrăm cu

$\begin{bmatrix} a_1 & b_2 \\ b_2 & a_2 \end{bmatrix} \Rightarrow$ Dacă $b_2 \to 0$ $\lambda \simeq a_1$ sau $\begin{bmatrix} a_{n-1} & b_n \\ b_n & a_n \end{bmatrix} \Rightarrow$ Dacă $b_n \to 0$, $\lambda \simeq a_n$